P-S-730B
June 30, 1976
SUPERSEDING
Fed. Spec. P-S-730A
May 15, 1964

FEDERAL SPECIFICATION

STAIN REMOVER, TABLEWARE, OXYGEN RELEASING TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a stain remover for use with aluminum, plastic, glass, or china tableware and coffee urns (see 6.1).

1.2 Classification.

1.2.1 Forms. The stain remover covered by this specification shall be of the following forms as specified (see 6.2).

Form 1 - Two-pound waterproof pouch.
Form 2 - Ten one-ounce water-soluble envelopes in a waterproof box.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-378 - Plastic Sheet and Strip (Polyolefin).
QQ-A-250/4 - Aluminum Alloy 2024, Plate and Sheet.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 7930

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:18Z
Check the source to verify that this is the current version before use.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

16 CFR 1500 - Federal Hazardous Substance Act.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 1331 - Surface and Interfacial Tension of Solutions of Surface-Active Agents.
D 2180 - Active Oxygen in Bleaching Compounds.
E 70 - pH of Aqueous Solutions with the Glass Electrode.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The stain remover shall be a clean, uniform, free-flowing granular or powdered material and shall be free of any objectionable odor in either the dry form or aqueous solution. It shall contain an oxygen-releasing agent, such as perborate or percarbonate, crystal growth inhibitor, surfactant and oxygen stabilizer.

3.1.1 Form 1. The stain remover shall be unit-packaged in 2-pound capacity vapor-proof pouches, as specified in 5.1.1.1. Each 2-pound pouch shall contain 32 ± 1.0 ounce of powdered or granular stain remover.

3.1.2 Form 2. The stain remover shall be unit packaged in premeasured water-soluble envelopes made of polyvinyl alcohol as specified in 5.1.1.2. Three 1.0-ounce packets shall contain an average of 1.0 ± 0.03 ounce of powdered or granular stain remover.

3.1.3 Excluded ingredients. The stain remover shall contain no caustic.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:18Z
Check the source to verify that this is the current version before use.

3.2 Chemical and physical requirements. The stain remover shall conform to the requirements as specified in table I.

TABLE I. Requirements

	Minimum	Maximum
Available oxygen, percent	5	-
Solution stability, active oxygen, percent	4.5	-
Soap content	-	None
pH value at 25°C (77°F), 1.0 percent solution	10.0	11.5
Chlorine releasing compounds	-	None
Oven stability, loss of original active oxygen, percent 1/	-	5
Insoluble matter, percent	-	2.0
Insoluble matter, empty water-soluble envelope, percent, form 2	-	2.0
Surface tension, 1.0 percent solution, dynes per centimeter	-	32

1/ The stain remover shall show no evidence of caking or lumping.

3.3 Stability in storage. The stain remover shall be stable, homogenous, and show no evidence of caking, lumping, leakage from its container or deteriorate in performance when stored for a period of not less than 6 months after receipt under general warehouse conditions at temperature below 110°F (43.4°C).

3.4 Metal. The stain remover shall show no corrosive effects on aluminum alloy tested as specified in 4.4.10.

3.5 Labeling. The containers shall be labeled in accordance with the Federal Hazardous Substances Act. Each pouch or box of stain remover shall be durably and legibly marked with the following information, precautions, and directions for use:

CAUTION: KEEP CONTAINER TIGHTLY CLOSED.
STORE IN A COOL DRY PLACE.
ALKALINE. Avoid contact with skin and eyes. Avoid breathing dust from powder. In case of contact, immediately flush skin or eyes with water for at least 15 minutes and get medical attention.

DIRECTIONS FOR USE: The manufacturer's recommended dilutions and "Directions For Use" shall be clearly shown on the label.

3.6 Water-vapor permeability of two-pound unit pouches. When tested as specified in 4.4.12, the bag shall have a water-vapor permeability of not more than 1.30 grams per 100 square inches in 24 hours (20.15 grams per square meter in 24 hours).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Examination of preparation for delivery. Packaging, packing and marking shall be examined to determine compliance with the requirements of section 5. Defects shall be scored in accordance with table II. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the lot offered for inspection at one time. Sampling shall be in accordance with MIL-STD-105 inspection level S-2, Acceptable Quality Level (AQL) 4.0 percent defective.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:18Z
Check the source to verify that this is the current version before use.

TABLE II. Examination of preparation for delivery

Examine	Defect
Contents	Not as specified.
Containers	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of container.
	Cushioning inadequate, improper or missing.

4.2.1.1 Examination for closure, waterproofing and banding of shipping containers. When shipping containers are required to comply with PPP-B-636, examination for defects in closure, waterproofing and banding shall be in accordance with the appendix of that specification.

4.2.1.2 Examination of the end item. Unit containers shall be inspected for defects as specified in table III. The sample unit shall be one unit container. The lot size shall be the number of unit containers in the lot. An inspection lot shall consist of all unit containers which are from the same container lot, which have been filled with stain remover from the same batch, and which are offered for delivery at the same time. Sampling and inspection shall be in accordance with MIL-STD-105, inspection level I, AQL 2.5 percent defective.

TABLE III. Examination of end item

Examine	Defect
Material	Not granular or powdered.
Workmanship	Not uniform; not free from foreign matter. Caked, contains lumps and constituents segregated.

4.2.1.3 Examination of net weight for the two-pound pouch. An examination of the net weight shall be as follows: The sample unit shall be one filled pouch. The lot size shall be expressed in two-pound pouches. The inspection level shall be S-3 and the AQL shall be 2.5 expressed as percent defective.

Examine	Defect 1/ 2/
Net weight for the 2-pound pouch.	Net weight more than one ounce under 32 ounces.
	Net weight more than one ounce over 32 ounces.

1/ Lot shall be rejected if sample data indicate a lot average net weight less than specified net weight.
2/ Weighing device shall have at least one-ounce increments of measurements.

4.3 Testing of the end item. The end item shall be tested as specified in table IV. Sampling shall be in accordance with MIL-STD-105, inspection level S-2. The lot shall be unacceptable if any sample unit fails any of the specified tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:18Z
Check the source to verify that this is the current version before use.

TABLE IV. Instructions for testing

Characteristic	Specification reference Requirement	Test method	Number determinations per unit composite	Results reported as Pass/Fail 1/	Numerically to nearest
Available oxygen	Table I	4.4.1	Average of 2	---	0.1 percent
Solution stability	Table I	4.4.2	Average of 2	---	0.1 percent
Soap content	Table I	4.4.3	1	X	
pH value	Table I	4.4.4	Average of 2	---	0.1 pH
Chlorine releasing compound	Table I	4.4.5	2	X	
Oven stability	Table I	4.4.6	Average of 2	---	0.1 percent
Insoluble matter	Table I	4.4.7	Average of 2	---	0.1 percent
Insoluble matter, empty water-soluble envelope	Table I	4.4.8	Average of 2	---	0.01 percent
Surface tension	Table I	4.4.9	Average of 2	---	1.0 dyne/cm
Effect on aluminum alloy	3.4	4.4.10	1	X	

1/ If failure is indicated, report either description of failure or numerical point of failure, as applicable.

4.4 Test methods.

4.4.1 Available oxygen. The available oxygen shall be determined in accordance with ASTM D 2180.

4.4.2 Solution stability. Dissolve 50.0 g of stain remover in 900 ml of distilled water at room temperature in a tared 1500-ml beaker. Add distilled water to bring total weight of solution to 1000 g. Heat the solution to 180°F (82.2°C). Mark the liquid level on the beaker. Heat at 180° ± 3°F (82.2 ± 1.6°C) for 3 hours, restoring the water lost by evaporation. The solution shall then be cooled to room temperature using an ice water bath and the solution weight adjusted to the original 1000 g. Take a 10.0 g sample of the solution and add 100 ml of 5 percent (by volume) sulfuric acid solution. Determine the available oxygen as specified in 4.4.1.

4.4.3 Qualitative test for soap. Dissolve 2 g of stain remover in 100 ml of distilled water. Acidify 5 ml of the filtrate with glacial acetic acid in test tube. The appearance of fatty acid precipitate or oily drops indicates the presence of soap.

4.4.4 pH value. Determine the pH at 25° ± 1°C (77° ± 1.8°F) of a solution containing 1.0 g of stain remover per 100 ml of distilled water in accordance with ASTM E 70.

4.4.5 Chlorine releasing compounds. Sprinkle a small quantity of stain remover on a strip of wet potassium iodide-starch paper. The appearance of blue-black color on the paper indicates the presence of active chlorine.

4.4.6 Oven stability. Weigh accurately to the nearest mg, a 4.0-g sample of stain remover into each of three 400-ml tared beakers. Place in an oven and maintain at 70° ± 1°C (158° ± 1.8°F) for 24 hours. Determine the available oxygen of these samples as specified in 4.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:18Z
Check the source to verify that this is the current version before use.

4.4.7 Insoluble matter in the stain remover. Dissolve approximately 5.0 g of stain remover accurately weighed to the nearest ± 1 mg in 100 ml of distilled water in a 250-ml beaker. Heat the solution to 180°F (82.2°C) and mark the liquid level on the beaker. Keep the solution at 180° ± 3°F (82.2° ± 1.6°C) for 15 minutes, replacing any water lost by evaporation. Filter through a tared Gooch crucible containing a fiber-glass filter. Wash the crucible with 100 ml of distilled water at 180°F (82.2°C). Dry the crucible at 105°C (221°F) for 4 hours. Cool to room temperature in a desiccator. Weigh the crucible and contents quickly after removal from desiccator and calculate the percentage of insoluble matter.

4.4.8 Insoluble matter in the empty plastic envelope. Dissolve one empty water-soluble plastic envelope, accurately weighed to the nearest 0.1 mg, in 500 ml of distilled water. Filter through a dried tared Gooch crucible containing a fiber-glass filter. Wash the crucible with hot distilled water at 70°C (158°F). Dry the crucible, with residue, at 105°C (221°F) for 4 hours. Cool to room temperature in a desiccator. Weigh the crucible and contents quickly after removal from desiccator and calculate the percentage of insoluble matter.

4.4.9 Surface tension. Completely dissolve 1.00 g of the stain remover in 80 ml of distilled water at 25° ± 1°C (77° ± 1.8°F). Determine the surface tension in accordance with ASTM D 1331, sections 1 to 7.

4.4.10 Effect on aluminum alloy. Polish, by buffing, a 1 cm x 8 cm panel of aluminum alloy conforming to QQ-A-250/4. Immerse the panel in a 2.0 percent by weight solution of stain remover for 2 hours at 140°F (60°C). After 2 hours, remove the panel and thoroughly rinse with distilled water, and dry. The panel shall show no evidence of corrosion or discoloration.

4.4.11 Water-soluble envelope product weight. Three packets, containing stain remover, shall be weighed individually to the nearest 0.01 oz. The product shall be removed from each packet and the packet shall be reweighed. The average net contents of three packets shall fall between the limits of 3.1.2.

4.4.12 Test for water vapor permeability. The sampling and testing for water vapor permeability shall be performed in accordance with the quality assurance provisions of L-P-378.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, C as specified (see 6.2).

5.1.1 Level A and B.

5.1.1.1 Pouches. Two pounds of stain remover shall be packaged in a water-vapor proof pouch commercially used for the product. The pouch shall have a water vapor permeability of not more than 1.30 grams per 100 square inches (20.15 grams per square meter in 24 hours) when tested as specified in 4.4.12.

5.1.1.2 Boxes. Ten one-ounce envelopes shall be packaged in a box conforming to style II, type D, class A of PPP-B-566. The interior of each box shall be lined with a minimum 0.0015-inch thick moisture-proof polyethylene bag. The bag shall be sealed to prevent accidental opening.

5.1.2 Level C. Stain remover shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Pouches. Thirty pouches specified in 5.1.1.1 shall be packed in a box conforming to style RSC-L, grade V3c of PPP-B-636. The box shall be closed, waterproofed, and banded in accordance with the appendix of PPP-B-636.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:18Z
Check the source to verify that this is the current version before use.

5.2.1.2 Boxes. Twenty-four boxes of stain remover as specified in 5.1.1.2 shall be packed in a box conforming to class weather resistant. The box shall be closed, waterproofed and banded in accordance with the appendix to PPP-B-636.

5.2.2 Level B.

5.2.2.1 Pouches. Thirty pouches specified in 5.1.1.1 shall be packed in a box conforming to style RSC-L, class domestic, grade 275 of PPP-B-636. The box shall be closed in accordance with the appendix to PPP-B-636.

5.2.2.2 Boxes. Twenty-four boxes of stain remover as specified in 5.1.1.2 shall be packed in a box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to PPP-B-636.

5.2.3 Level C. Stain remover in quantities as specified (see 6.2) and packaged as specified in 5.1 shall be packed in containers to assure carrier acceptance and safe arrival at destination. Container shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking.

5.3.1 Labeling and marking. The labeling and marking shall comply with the Federal Hazardous Substances Act and Regulations Promulgated thereunder.

5.3.2 Civil agencies. Marking of the interior packages and external shipping container shall be in accordance with 3.5 and Fed. Std. No. 123 (see 6.2).

5.3.3 Military agencies. Marking of the interior packages and external shipping containers shall be in accordance with MIL-STD-129 (see 6.2).

5.3.4 Special marking.

5.3.4.1 Each box of plastic envelopes. Each box of the ten envelopes must indicate that plastic envelopes are water-soluble.

6. NOTES

6.1 Intended use. The stain remover covered by this specification is intended for use in hard or soft water for removal of coffee, tea and other absorbed food stains from plastic, glass, china, stainless or aluminum tableware. When used on stainless flatware it removes bluish haze and preserves its new look.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size of container required (see 5.1 and 5.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Quantities required (see 5.2.3).
(e) Marking document required (see 5.3).

MILITARY INTEREST:

DOD has waived coordination of this document until further notice.

Civil Agency Coordinating Activities:

GSA-FSS
HEW-NIH
VA-DMS

Preparing Activity: GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:18Z
Check the source to verify that this is the current version before use.